Project Design Phase

Problem – Solution Fit Template

Date	27 September 2026
Team ID	
Project Name	Metro Ticket Generating System Using ServiceNow
Maximum Marks	2 Marks

Problem – Solution Fit Template:

The Problem-Solution Fit means that a real problem has been identified for the customer and that the solution actually solves it. The canvas below applies it to the Metro Ticket Generating System.

Purpose / Vision: Replace repetitive manual ticket booking, fare calculation and status updates with a centralized, automated ServiceNow application.

1. CUSTOMER SEGMENT(S) [CS]

- Daily metro commuters
- Occasional passengers and travellers, including those going to the Airport
- Ticketing counter staff and administrators who record bookings and maintain station and fare data

6. CUSTOMER CONSTRAINTS [CC]

- Limited time at ticket counters, especially in busy periods
- Uncertainty about the correct fare
- Need for a simple process that does not require repeated entry of details
- Prototype limits: no payment gateway, no link to a real metro operator, simplified fares

5. AVAILABLE SOLUTIONS [AS]

- Ticket counters and separate ticketing or record-keeping processes

Pros: familiar; staff can help in person.

Cons: waiting time, manual entry, risk of fare errors, records spread across places, manual status updates.

(General limitations of some conventional processes, not of every metro system.)

2. JOBS-TO-BE-DONE / PROBLEMS [J&P]

- Passenger: choose source and destination, know the fare, get a ticket with an ID and confirmation
- Staff: record bookings accurately, keep station and fare data up to date, track booking status

Problem: these jobs rely on repetitive manual steps.

9. PROBLEM ROOT CAUSE [RC]

- Fare calculation depends on manual work or separate processes
- Booking records and status are not managed in one centralized place
- No automated rule or workflow to calculate fares, identify tickets and update status consistently

7. BEHAVIOUR [BE]

- Passengers go to the counter, ask for the fare and wait to be served
- Staff enter journey details by hand and update the booking status manually
- Staff look in different records to find booking information

3. TRIGGERS [TR]

- Need to make a journey
- Seeing a long queue or being unsure of the fare
- A new booking being created (starts fare calculation and the status flow)

4. EMOTIONS: BEFORE / AFTER [EM]

Before: rushed, uncertain about the fare, frustrated by waiting.

After: confident, in control, assured that the booking is confirmed and identifiable by its ticket number.

10. YOUR SOLUTION [SL]

A ServiceNow scoped application with Station, Fare Rule and Tickets Booking tables.

- A Business Rule calculates the fare (₹10 base + ₹5 per station); Central to Airport, one passenger = ₹20 in testing
- Each booking has a unique ticket number
- Flow Designer flow "Metro Booking Confirmation" sets the status to Confirmed on booking creation
- QR Code Data field prepared for future digital tickets

8.1 ONLINE [CH]

- Digital booking records and forms in ServiceNow
- Future: passenger self-service portal, email notifications, QR-code tickets

8.2 OFFLINE [CH]

- Ticket counters and station staff
- In-person fare enquiries and paper-based or manual records

Note: This is a prototype. QR-code generation, notifications, payments and a passenger self-service interface are proposed extensions, not completed features.

References:

1. https://www.ideahackers.network/problem-solution-fit-canvas/
2. https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe